

Classification d'attributs faciaux — pipeline MLOps ...

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https://github.com/omdrift/mlops_face_attribute_classification

Objectif

Objectif : Construire un pipeline complet pour classer des attributs faciaux, de l'ingestion à l'inférence.

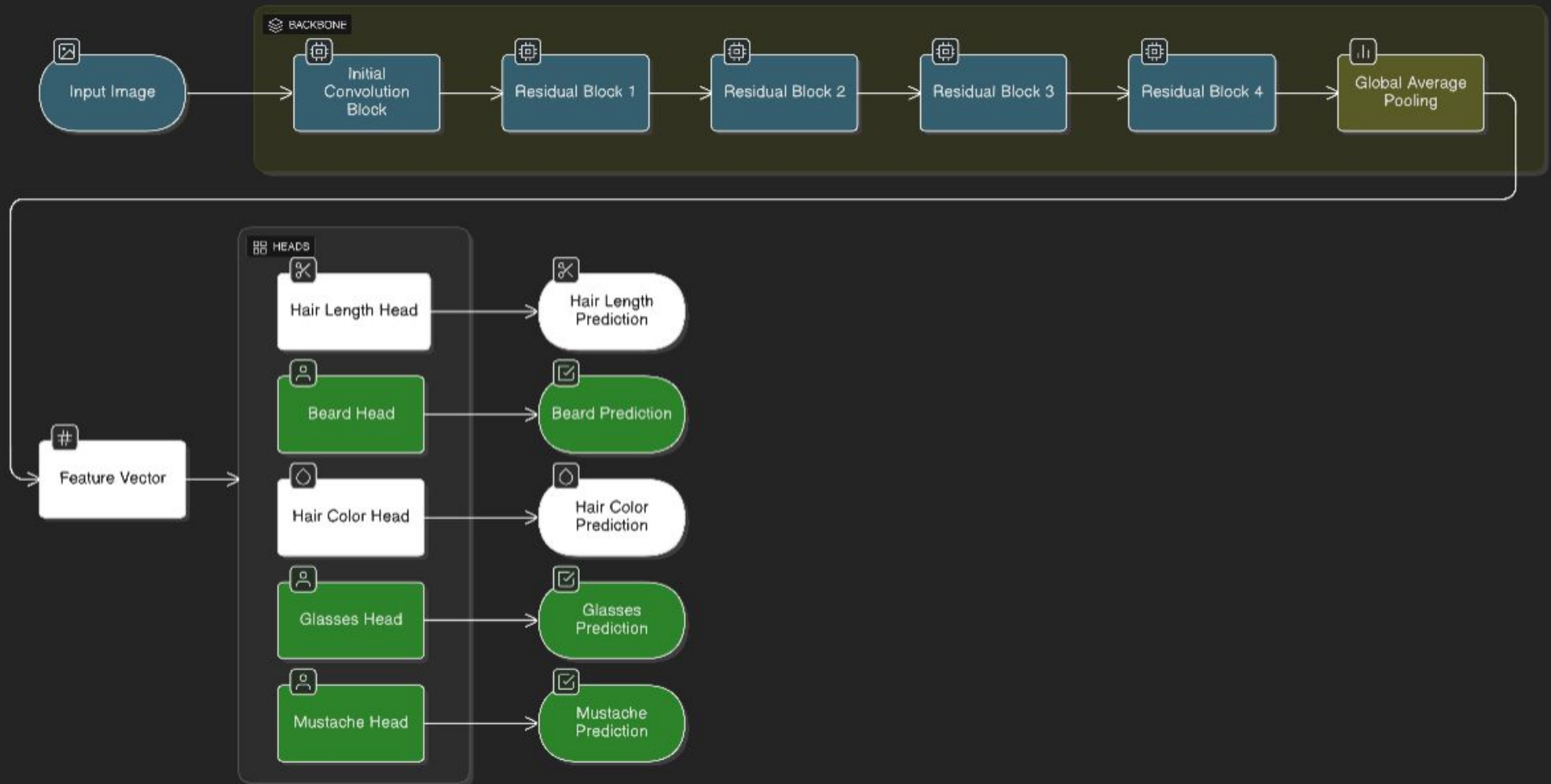
- Périmètre actuel : préparation des données, entraînement, suivi d'expériences, orchestration, déploiement.

Organisation du dépôt

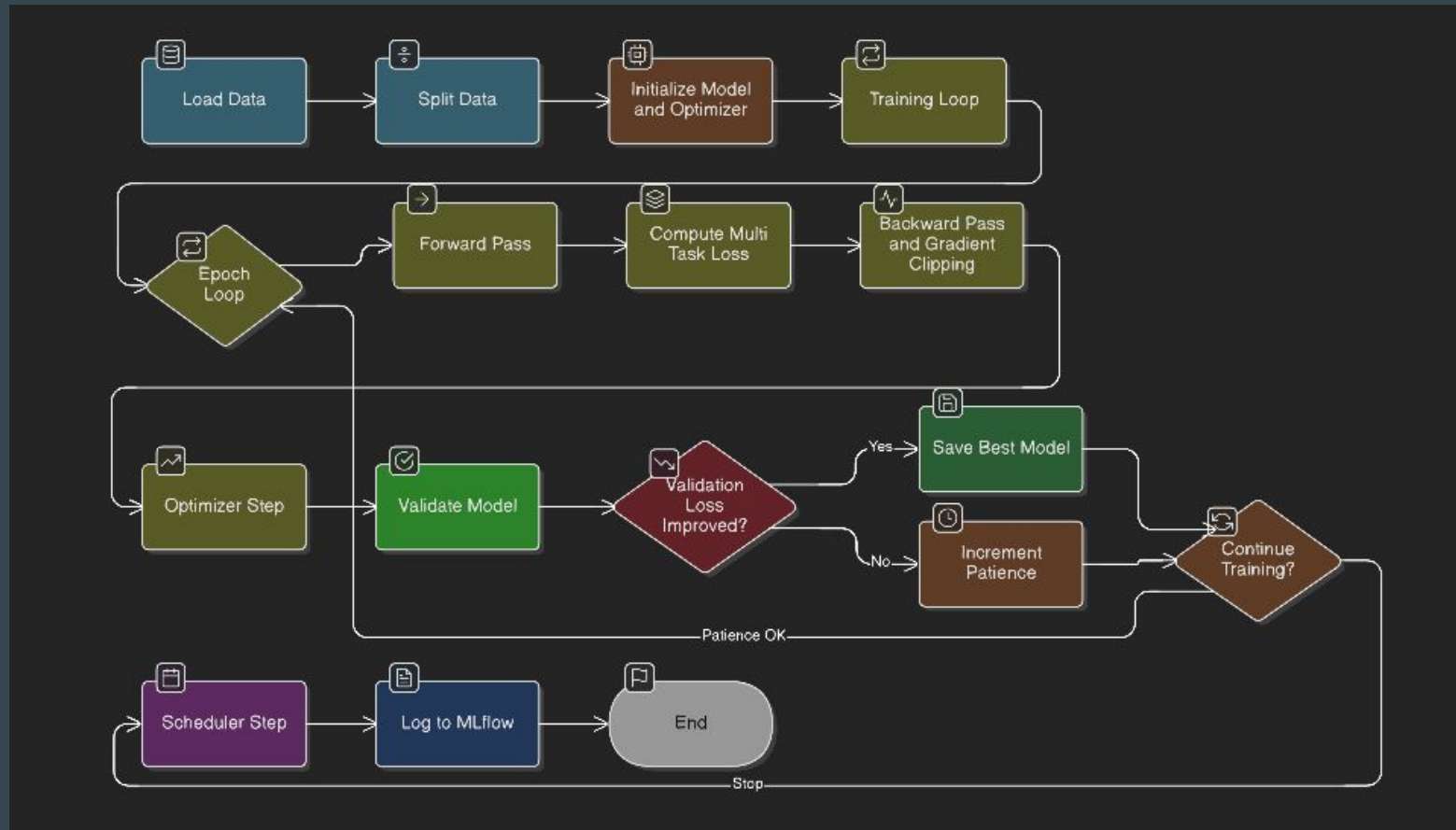
Notebooks : exploration, entraînement, évaluation.

- Code : src/ :préprocessing, modèle, utilitaires, tests/ :tests unitaires.
- Orchestration : airflow/ (DAGs pour pipeline).
- Données : data/ (gérées via DVC :`dvc.yaml`,`dvc.lock`).
- Expérimentations : MLflow et métriques .
- Déploiement : scripts/templates pour servir le modèle .
- Autres artefacts :`accuracy\_curves.png`,`training\_curves.png`,`model\_summary.txt`,`params.yaml`,`requirements.txt`,`pyproject.toml`.

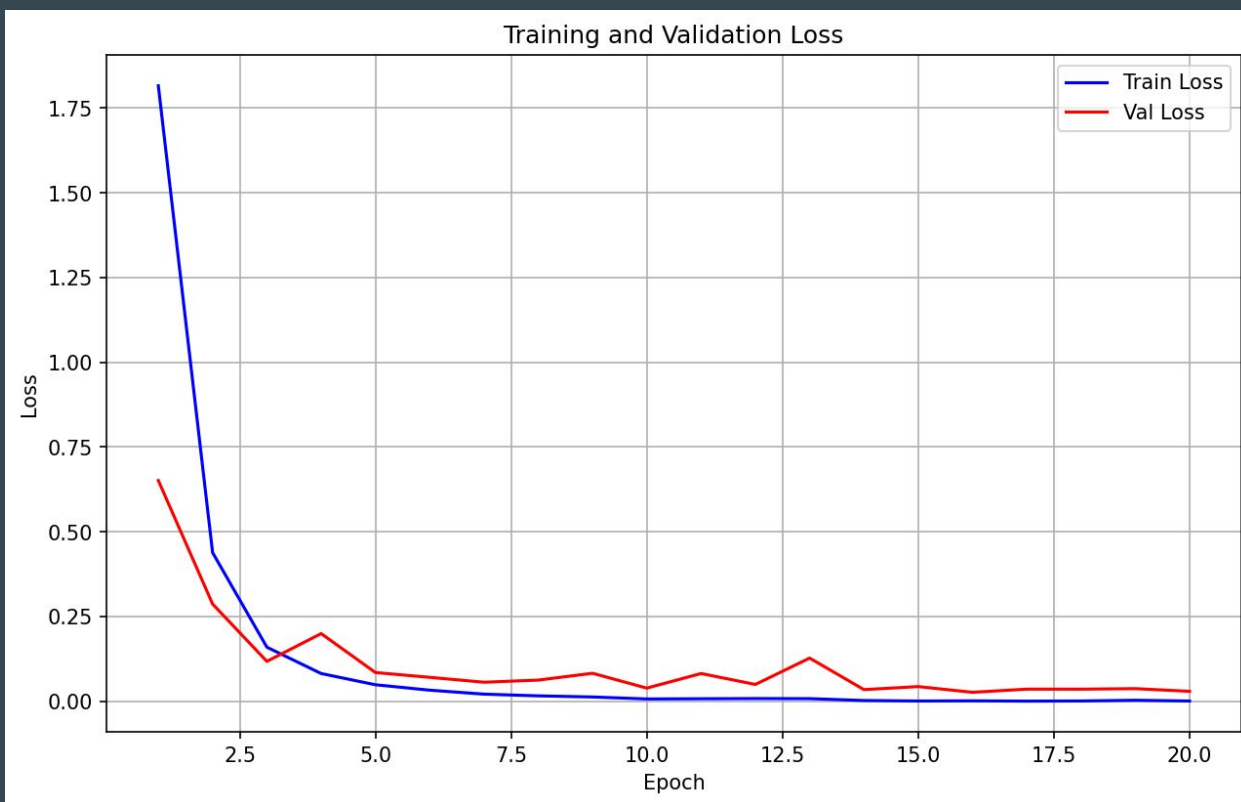
Model Architecture



Training

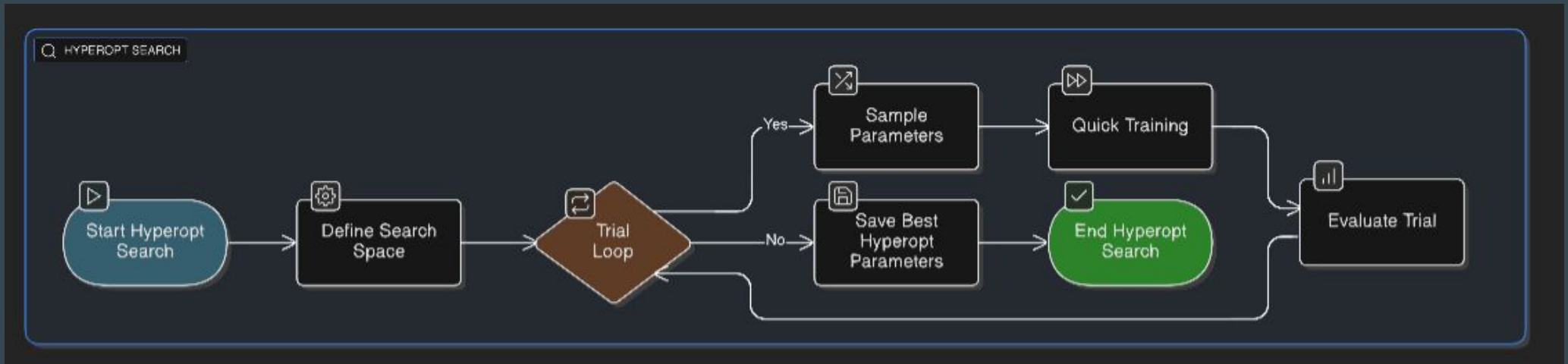


Loss curve

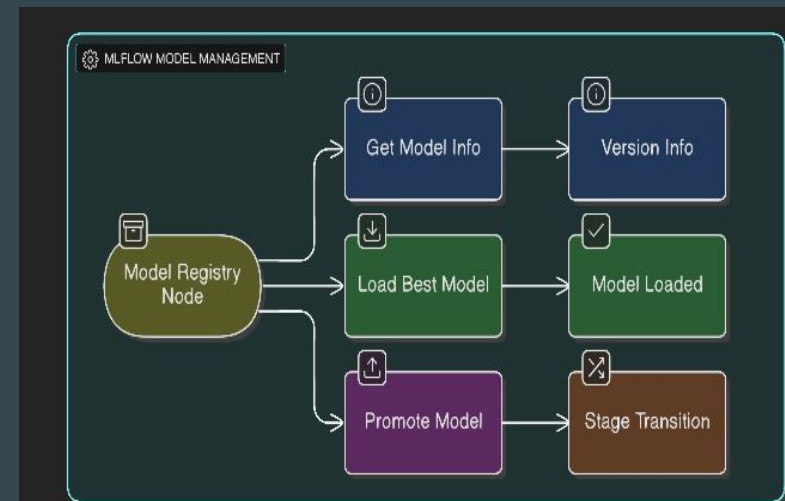


- optimizer: Adamw
- Early stopping
- Learning Rate Scheduler
- Hyperparam opti via hyperopt

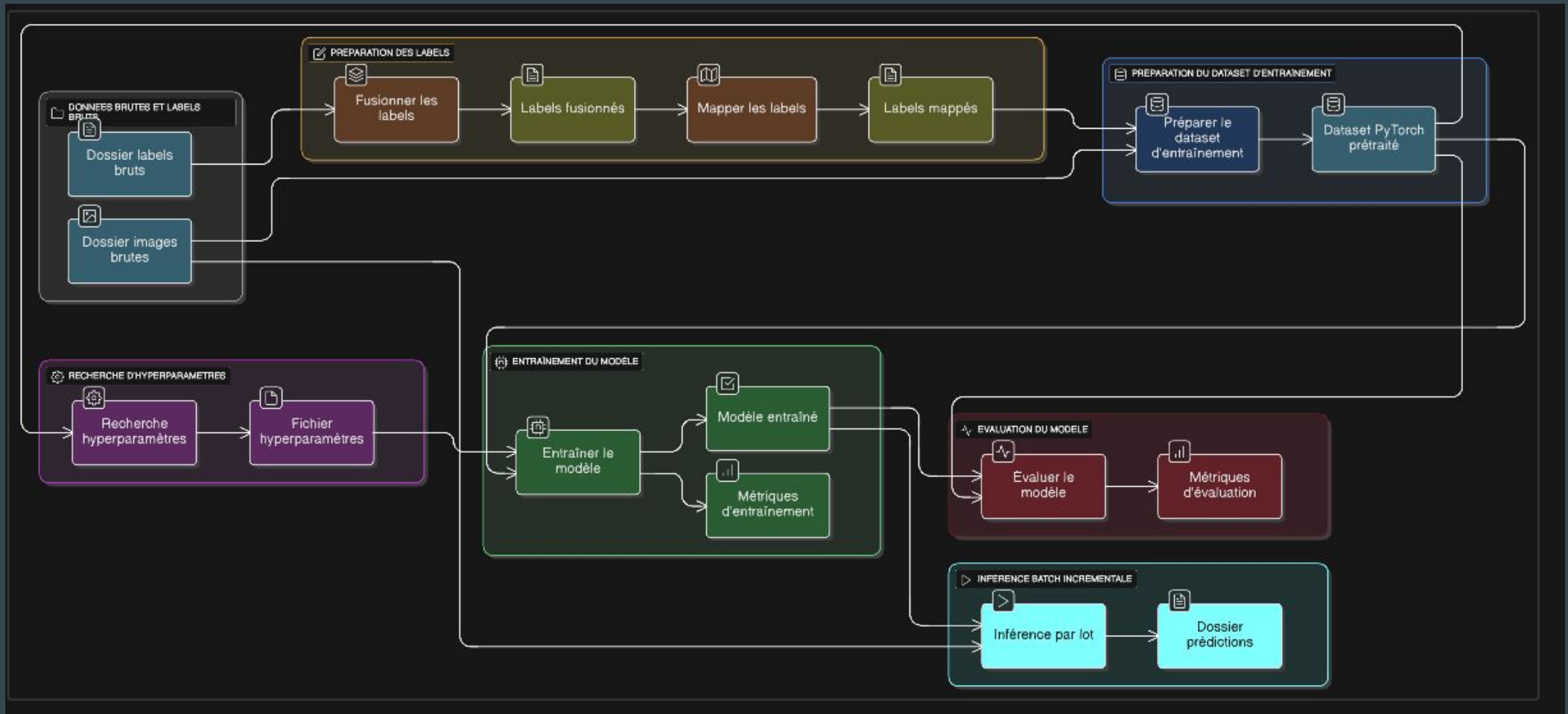
Hyperopt



Mlflow



Pipeline automatisée avec dvc



DAGs

All 3

Active 2

Paused 1

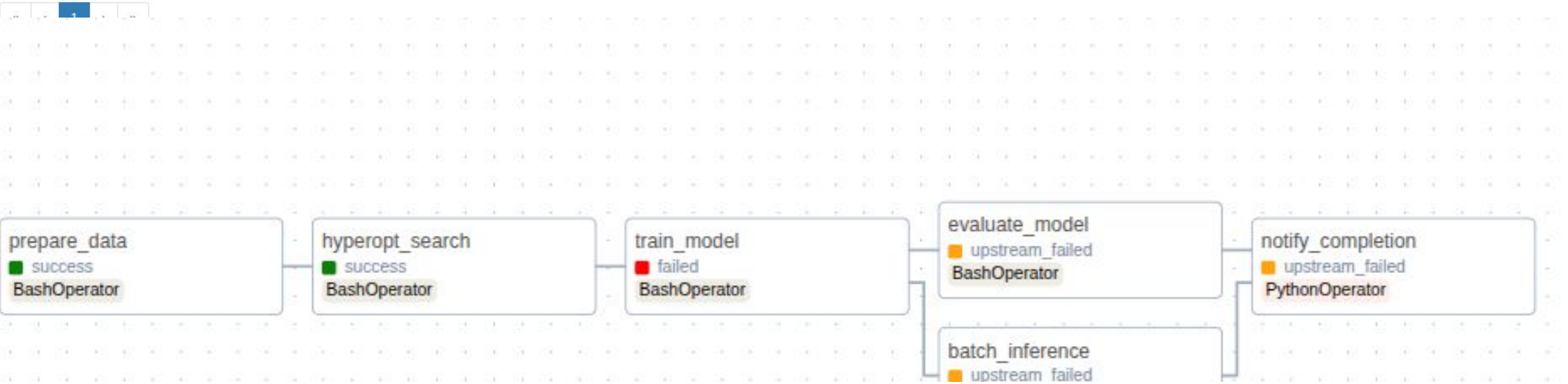
Running 0

Failed 1

Filter DAGs by tag

Search DAGs

DAG	Owner	Runs	Schedule	Last Run	Next Run	Recent Tasks
ml_inference_only inference ml	mlops-team	2	@daily	2025-12-04, 01:12:20	2025-12-04, 00:00:00	1
ml_pipeline dvc face-attributes ml	mlops-team	1 1	@weekly	2025-12-04, 01:11:46	2025-11-30, 00:00:00	2 1 3



Github Actions

CI Pipeline:

Trigger: on push/PR

Cache pip pour accélérer

Codecov integration

DVC Validation:

Trigger: dvc.yaml changes

Validation syntax only

No data required

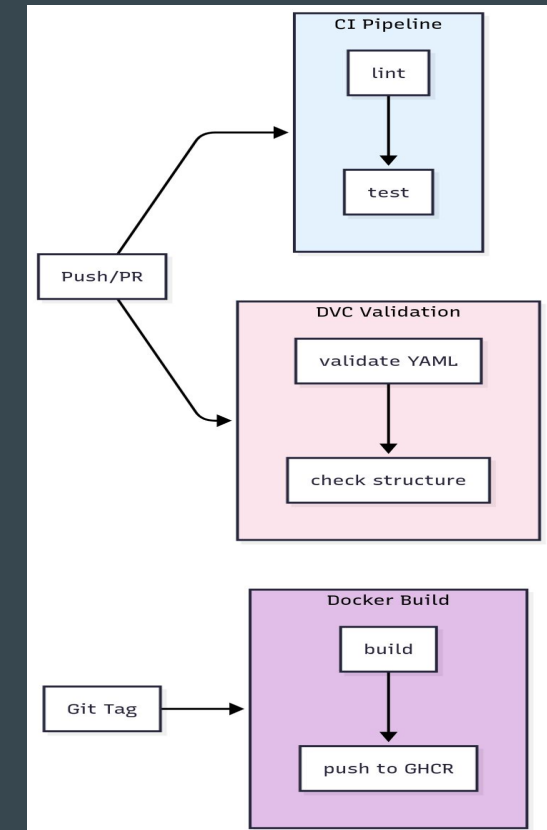
Docker Build:

Trigger: git tag v*.*.*

Registry: GHCR

Multi-stage caching

Automated versioning



Application dockerise

